



Habib University
shaping futures

WATER : SCIENCE, SOCIETY & POLICY

ENVS/SDP 251 L1

Spring Semester 2025

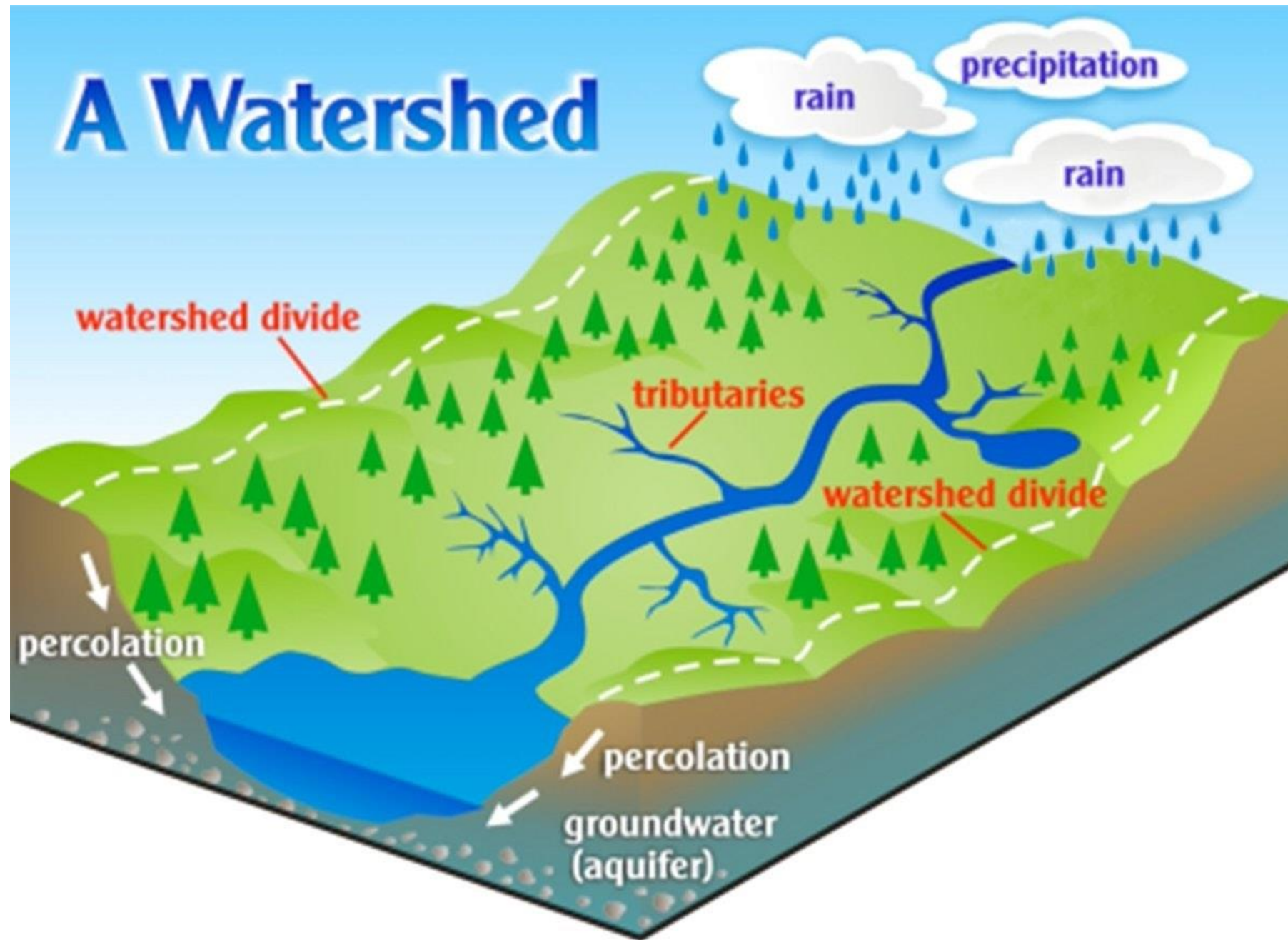
Deadlines

- Water Journal (Jan 21) Done
- Project proposal (Jan 28) Approaching
- Quiz 01 (Feb 13)

Research Project (30%)

- Group activity (Groups of 3)
- Students review literature and conduct field work to understand local water related problems and propose possible solutions.

	Grade % (Total 30%)
Proposal	2.5
Introduction	2.5
Literature review	2.5
Primary research	7.5
Presentation	5
Final Report	10



Watershed balance

- the equation to express the change in surface water storage
 $\Delta S = P - ET - SF - \Delta GW$

Where:

P: Precipitation (input to the watershed).

ET: Evapotranspiration (loss of water through evaporation and transpiration).

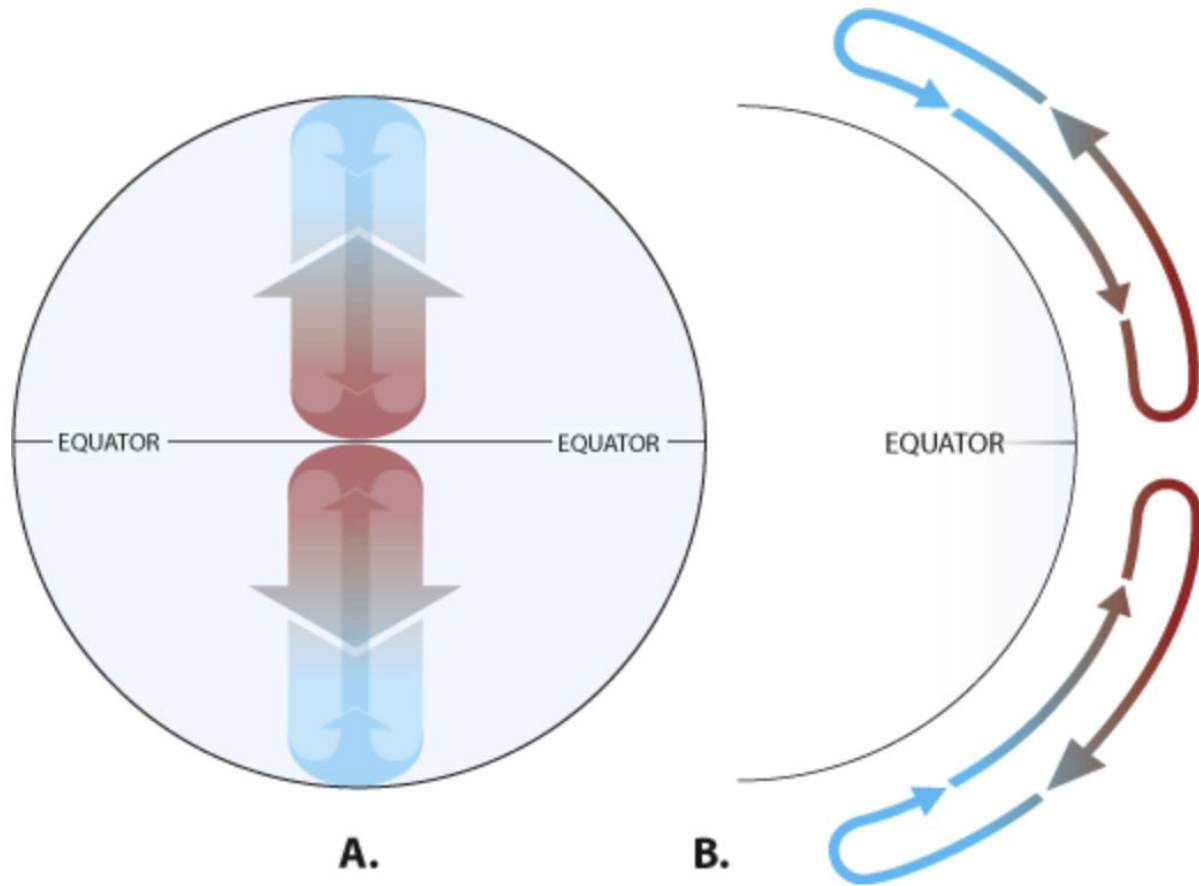
SF: Surface Flow (runoff leaving the watershed).

ΔGW : Net change in groundwater storage (groundwater inflow minus groundwater outflow).

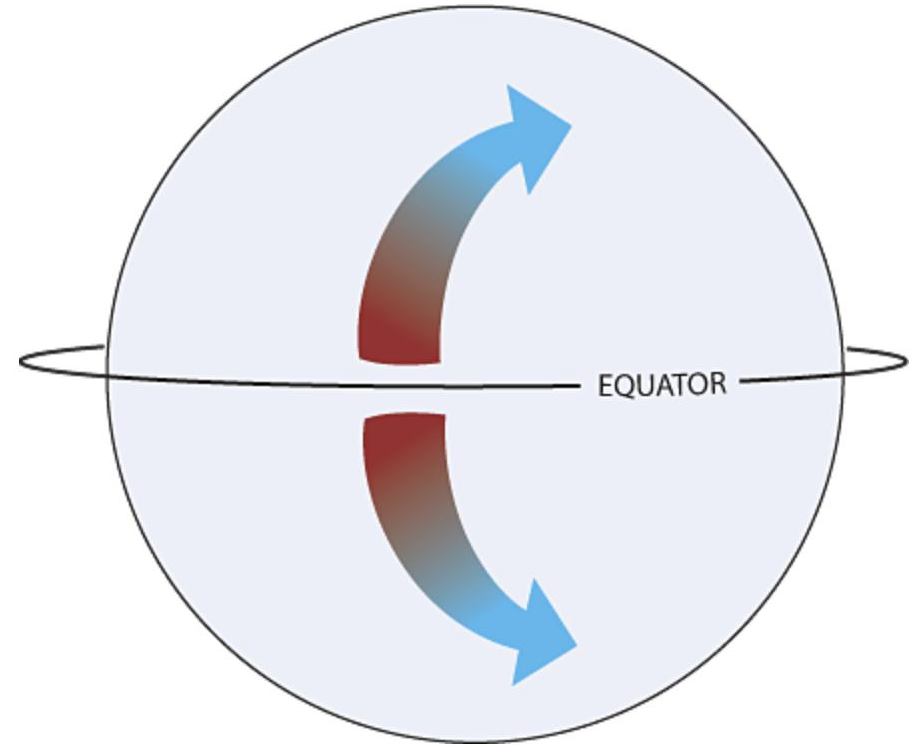
ΔS : Change in surface water storage (e.g., reservoirs, lakes, soil moisture).

- $\Delta GW = GW_{in} - GW_{out}$ (net change in groundwater storage).

Wind Patterns



Stationary earth



Rotating earth

Monsoon Rains in South Asia:

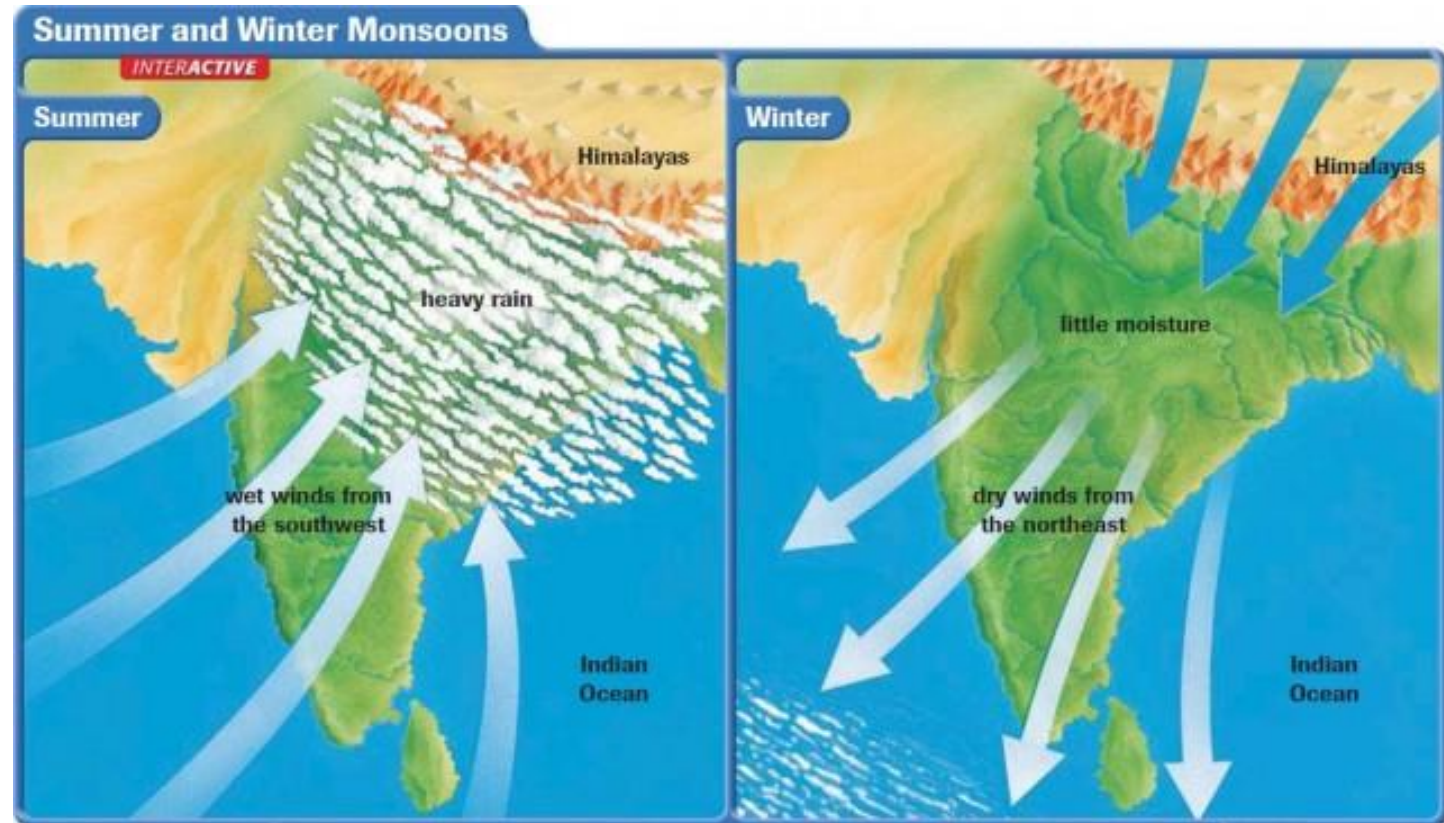
Region:

South Asia, including countries like Pakistan, India, Bangladesh, and Nepal.

Precipitation Pattern:

The South Asian monsoon is characterized by seasonal winds that bring heavy rainfall during the summer months. This pattern results in a wet monsoon season (June-September) and a dry season (October-May).

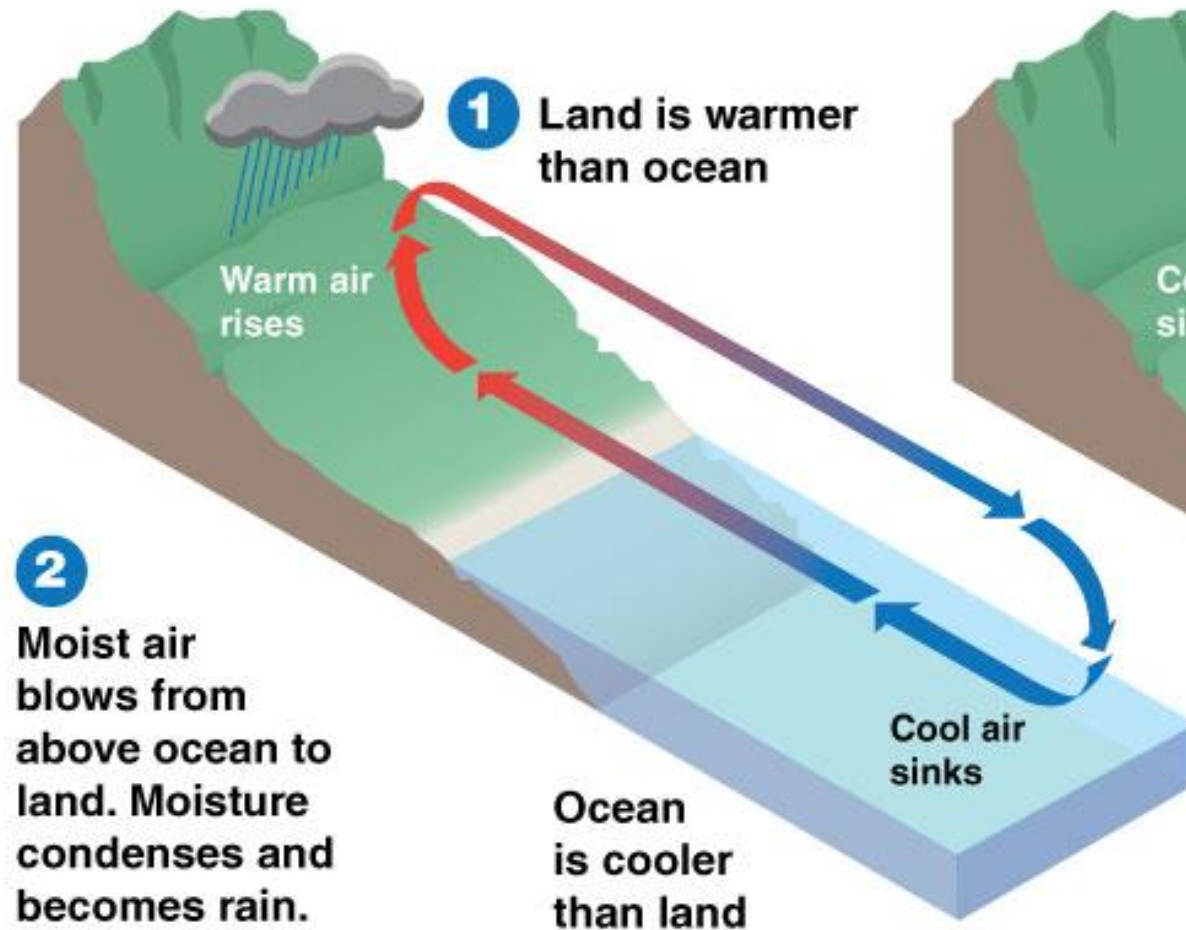
Impact: The monsoon rains are essential for agriculture in the region, providing water for crops. However, excessive or poorly distributed monsoon rains can lead to flooding, landslides, and damage to infrastructure.



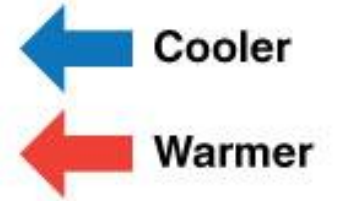
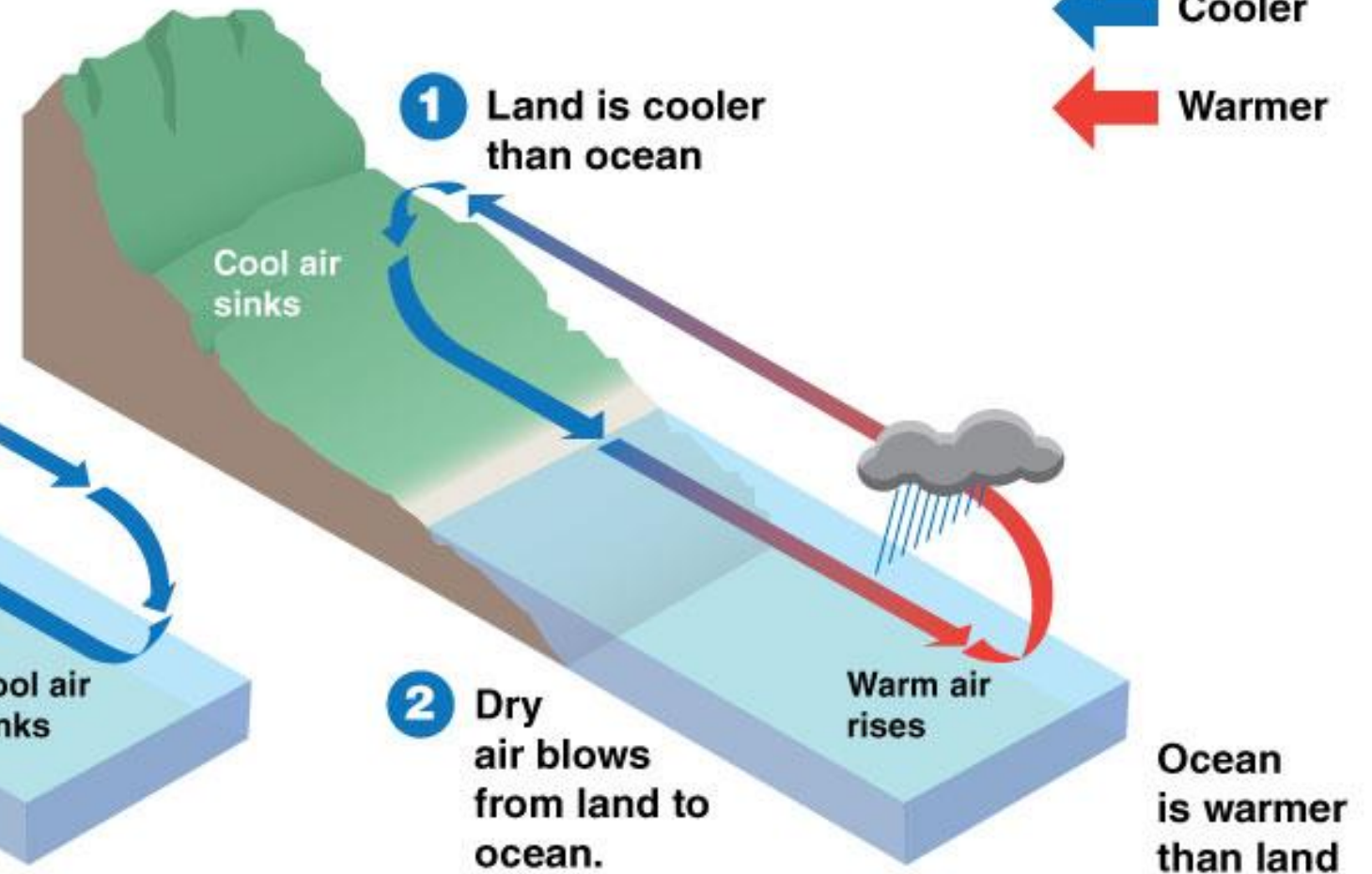
Source: geography.name

How a monsoon works

Spring/Summer



Winter



Rain Shadow Effect in the American West:

Region: Western United States, including areas like the Pacific Northwest and the Great Basin.

Precipitation Pattern: Mountains along the western coast, such as the Sierra Nevada, block moist air from the Pacific Ocean. As a result, the windward side of the mountains receives heavy rainfall, while the leeward side (rain shadow) is dry.

Impact: The rain shadow effect creates stark differences in precipitation, with lush forests on the windward side and arid regions on the leeward side. This influences agriculture, water resources, and ecosystems.



Source: montananaturalist.org

Drought in the Sahel Region:

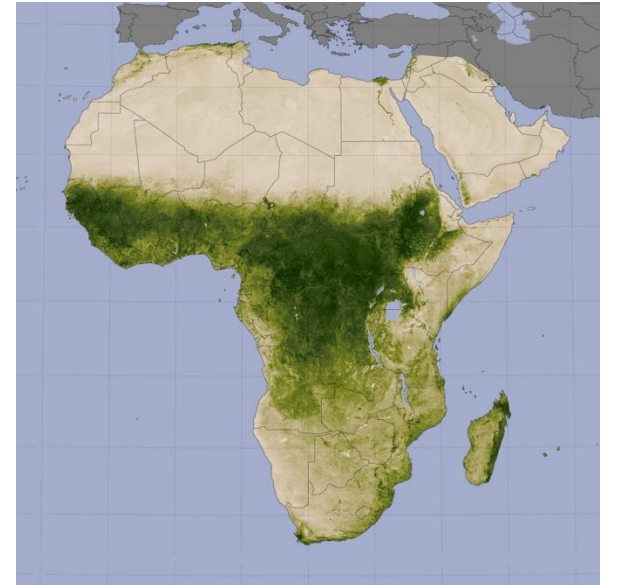
Region: Sahel region of Africa, including countries like Mali, Niger, Chad, and Sudan.

Precipitation Pattern: The Sahel experiences highly variable rainfall patterns. In recent decades, there have been periods of prolonged drought.

Impact: Droughts in the Sahel can lead to food shortages, water scarcity, and conflicts over resources. It has significant implications for food security and livelihoods.



Source: Aljazeera



Source: NASA.gov